

Disproof of Conjecture 00000007964: binary linear quasi-perfect codes exist for every length $n \geq 2$

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Abstract

Conjecture 00000007964 claims, among other things, that binary linear quasi-perfect codes of length n exist if and only if $n = 2^k - 1$ or n lies in a finite exceptional list. This is false. The conjecture defines quasi-perfect codes as codes whose covering radius is one more than the packing radius $e = \lfloor (d-1)/2 \rfloor$ and which have a “tight uniform shell distribution”. We treat two readings: the standard one ($\rho = e + 1$ only), and the conjecture’s stronger notion, where in addition any two words at the same distance from the code have equally many codewords in every shell. Under *both* readings such codes exist for every length $n \geq 2$ and for no length $n \leq 1$. The witnesses are not degenerate: for every $n \geq 3$ with $n + 1$ not a power of two, the shortened Hamming code of length n is a quasi-perfect $[n, n - \lceil \log_2(n+1) \rceil, 3]$ code. For every $m \geq 2$, the extended Hamming code of length 2^m is quasi-perfect with $d = 4$. Both the extended Hamming codes and the once-shortened Hamming codes of length $2^m - 2$ ($d = 3$) have uniform shell distributions. The set of lengths outside $\{2^k - 1\}$ is infinite, so no finite exceptional list can rescue the clause. All three code families, the full length spectrum and the refutation of the clause are formalized in Lean 4 (core library only). The Lean refutation covers the uniform-shell notion (via even-weight codes), the restrictions to $d \geq 3$ or $d \geq 4$, and the non-strict reading.

1 The conjecture and the clause we refute

The English statement reads:

Definition: The classification extension of perfect codes: nontrivial q -ary perfect codes (covering radius 1, balls fill the space) have known types the Hamming codes and the binary/ternary Golay codes; define relaxed quasi-perfect codes (Q-uniform ball measures) with tight uniform shell distribution at covering radius $r + 1$. *Conjecture:* The gap constant of the radius spectrum of quasi-perfect codes is at most 2 (covering efficiency at least $1/2$); and binary linear quasi-perfect codes of length n exist if and only if n is of type $2^k - 1$ or in a finite exceptional list (the Levenshtein set), completely enumerating the spectrum.

The Chinese version says the same thing. The conjecture is a conjunction. Its first conjunct (“gap constant of the radius spectrum”) is never defined, and we do not discuss it. Refuting the second conjunct suffices:

$$\begin{aligned} & \exists \text{ finite } L \subset \mathbb{N} \quad \forall n : \\ & (\exists \text{ binary linear quasi-perfect code of length } n) \iff (\exists k : n = 2^k - 1) \vee n \in L. \end{aligned} \tag{1}$$

Definition 1. A *binary linear code* of length n is an $\mathbb{F}_2$ -subspace $C \subseteq \mathbb{F}_2^n$ with $|C| \geq 2$. Its *minimum distance* is $d = \min\{d(x, y) : x \neq y \in C\}$ (Hamming distance), its *packing radius* is $e = \lfloor (d-1)/2 \rfloor$, and its *covering radius* is $\rho = \max_{x \in \mathbb{F}_2^n} \min_{c \in C} d(x, c)$. The balls of radius e around codewords are disjoint and those of radius ρ cover $\mathbb{F}_2^n$. Hence $\rho \geq e$, and C is *perfect* if $\rho = e$. C is *quasi-perfect* if $\rho = e + 1$.

This is the standard notion (for example, MacWilliams and Sloane). The conjecture's own definition is *stronger*. Its preamble asks for a “tight uniform shell distribution at covering radius $r + 1$ ”, where r is the packing radius. We formalize this as follows.

Definition 2. For $x \in \mathbb{F}_2^n$ and $j \geq 0$ let $S_j(x) = \{c \in C : d(x, c) = j\}$ be the j -th shell around x . The code C is *shell-uniform* if $|S_j(x)| = |S_j(y)|$ for all j whenever $d(x, C) = d(y, C)$. A *uniform-shell quasi-perfect* code is a quasi-perfect code that is shell-uniform.

This is the strongest reasonable reading: it constrains every shell, not only the shells of radius e and $e + 1$ that enter the classical notion of a uniformly packed code. We refute the clause under both the standard and the uniform-shell reading (Section 6 and Corollary 13). Section 8 covers the remaining readings: the non-strict version $\rho \leq e + 1$, a “nontriviality” demand $d \geq 3$ or $d \geq 4$, restrictions on k , and the choice of the finite list.

2 Syndromes

We use the bitwise exclusive or $a \oplus b$ of nonnegative integers. It is addition in the $\mathbb{F}_2$ -vector space of finitely supported bit strings. In particular it is associative and commutative, $a \oplus a = 0$, $a \oplus 0 = a$, and $a \oplus b = 0 \iff a = b$.

Fix labels $\ell_1, \dots, \ell_n \in \mathbb{N}$, one per coordinate. The *syndrome* of $x \in \mathbb{F}_2^n$ is

$$\sigma(x) = \bigoplus_{j: x_j=1} \ell_j.$$

Then $\sigma(x + y) = \sigma(x) \oplus \sigma(y)$, so $\ker \sigma$ is a linear code. This is just the code with parity-check matrix whose j -th column is the binary expansion of ℓ_j . Write e_j for the j -th unit vector.

Lemma 3. *Suppose the labels are pairwise distinct.*

1. *If $\text{wt}(x) = 0$ then $\sigma(x) = 0$. If $\text{wt}(x) = 1$, say $x = e_j$, then $\sigma(x) = \ell_j$. If $\text{wt}(x) = 2$ then $\sigma(x) \neq 0$.*
2. *If all $\ell_j < 2^m$ then $\sigma(x) < 2^m$ for every x .*
3. *For $x \in \mathbb{F}_2^n$ and y with $\sigma(y) = \sigma(x)$, the word $c = x + y$ lies in $\ker \sigma$ and $d(x, c) = \text{wt}(y)$.*

Proof. (1) $\sigma(e_i + e_j) = \ell_i \oplus \ell_j \neq 0$ for $\ell_i \neq \ell_j$. (2) XOR does not create bits above position $m - 1$. (3) $\sigma(x + y) = \sigma(x) \oplus \sigma(y) = 0$ and $x + c = y$. $\square$

Lemma 4. *If $t = 2^k \leq s < 2^{k+1}$ then $s \oplus t = s - t < t$.*

Proof. Bit k is the leading bit of s . XOR with 2^k clears it and leaves the lower bits unchanged. $\square$

3 Shortened Hamming codes: $d = 3$, every length $n \geq 3$ with $n + 1 \neq 2^k$

For $n \geq 1$ let $\text{SH}(n) = \ker \sigma \subseteq \mathbb{F}_2^n$ with labels $\ell_j = j$ ($j = 1, \dots, n$). For $n = 2^m - 1$ this is the Hamming code. For other n it is a shortened Hamming code.

Theorem 5. *Let $n \geq 3$ and suppose $n + 1$ is not a power of 2. Then $\text{SH}(n)$ is a binary linear code with minimum distance $d = 3$ (so $e = 1$) and covering radius $\rho = 2$. It is quasi-perfect.*

Proof. Minimum distance. Let $c \neq 0$ be a codeword. $\text{wt}(c) \neq 1$, because $\sigma(e_j) = j \neq 0$. $\text{wt}(c) \neq 2$ by Lemma 3(1). So $\text{wt}(c) \geq 3$. Since $1 \oplus 2 \oplus 3 = 0$, $e_1 + e_2 + e_3$ is a codeword of weight 3. For a linear code the minimum distance is the least nonzero weight, so $d = 3$.

Covering radius ≤ 2 . Let $k = \lfloor \log_2 n \rfloor$ and $t = 2^k$, so $t \leq n < 2t$. Let $x \in \mathbb{F}_2^n$ and $s = \sigma(x)$. By Lemma 3(2) with $m = k + 1$ we have $s < 2t$. We find y with $\text{wt}(y) \leq 2$ and $\sigma(y) = s$:

- if $s = 0$, take $y = 0$;
- if $1 \leq s \leq n$, take $y = e_s$;
- if $n < s < 2t$, put $b = s \oplus t$. By Lemma 4, $b < t \leq n$. Also $b \neq 0$, since otherwise $s = t \leq n$. So $y = e_t + e_b$ is a word of weight 2 with $\sigma(y) = t \oplus b = s$.

By Lemma 3(3), $c = x + y$ is a codeword with $d(x, c) \leq 2$.

Covering radius ≥ 2 . Since $n < 2t$ and $n + 1 \neq 2t$, we have $n \leq 2t - 2$. Put $s^* = 2t - 1$, so $n < s^* < 2t$. By the previous paragraph there is y^* with $\sigma(y^*) = s^*$. For every codeword c , $\sigma(y^* + c) = s^*$. Since $s^* \neq 0$ and $s^* \notin \{1, \dots, n\}$, Lemma 3(1) gives $\text{wt}(y^* + c) \notin \{0, 1\}$. So $d(y^*, c) \geq 2$ for all c , and $\rho = 2 = e + 1$. $\square$

For $n = 2^m - 1$ the same argument gives $\rho = 1 = e$: every $s < 2^m$ is a label or 0. So $\text{SH}(2^m - 1)$ is the perfect Hamming code and not quasi-perfect. The Lean file proves this as a sanity check (`hamming_not_QP`). The dimension of $\text{SH}(n)$ is $n - \lceil \log_2(n + 1) \rceil$, because the labels $1, 2, 4, \dots, 2^k \leq n$ make σ surjective onto $\{0, \dots, 2^{k+1} - 1\}$. This is not needed for the disproof.

4 Extended Hamming codes: $d = 4$, lengths 2^m

For $m \geq 2$ let $N = 2^m$. Label the coordinates $0, 1, \dots, N - 1$ and let

$$\text{EH}(m) = \{x \in \mathbb{F}_2^N : \sigma(x) = 0, \text{ wt}(x) \text{ even}\}.$$

This is the extended Hamming code. It is linear because σ and the parity $\text{wt}(x) \bmod 2$ are both linear.

Theorem 6. *$\text{EH}(m)$ has $d = 4$ (so $e = 1$) and covering radius 2, so it is quasi-perfect. Moreover, every word at distance 2 from the code has exactly $N/2$ codewords at distance 2. Every word at distance 1 has exactly one codeword at distance 1 and none at distance 2.*

Proof. A nonzero codeword has even weight, and weight 2 is excluded by Lemma 3(1). Since $0 \oplus 1 \oplus 2 \oplus 3 = 0$, $e_0 + e_1 + e_2 + e_3$ is a codeword of weight 4. So $d = 4$.

Let $x \in \mathbb{F}_2^N$ and $s = \sigma(x) < N$. If $\text{wt}(x)$ is odd, $y = e_s$ has syndrome s and odd weight. If $\text{wt}(x)$ is even and $s \neq 0$, $y = e_0 + e_s$ has syndrome s and even weight. If $\text{wt}(x)$ is even and $s = 0$,

x is a codeword. In each case $c = x + y$ is a codeword with $d(x, c) \leq 2$. For $x = e_0 + e_1$ and any codeword c , $x + c$ has syndrome $1 \neq 0$ and even weight, so $d(x, c) \geq 2$. Hence $\rho = 2$.

Shells: if $d(x, C) = 2$ then $\text{wt}(x)$ is even and $s = \sigma(x) \neq 0$. The codewords at distance 2 are the $x + e_a + e_b$ with $a \oplus b = s$, and there are exactly $N/2$ unordered pairs $\{a, a \oplus s\}$. If $d(x, C) = 1$, the codeword at distance 1 is unique, because $d = 4 > 2$. By parity no codeword is at even distance from x . $\square$

The shell numbers depend only on the distance to the code. So $\text{EH}(m)$ is uniformly packed: its shell distribution at the covering radius $e + 1$ is as uniform as it can be.

5 Even-weight and repetition codes

Proposition 7. *For $n \geq 2$ the even-weight code $\text{EW}(n) = \{x : \text{wt}(x) \text{ even}\}$ has $d = 2$, $e = 0$ and $\rho = 1$, so it is quasi-perfect. Every word not in the code has exactly n codewords at distance 1.*

Proof. Nonzero codewords have weight ≥ 2 , and $e_1 + e_2$ is a codeword. A word of odd weight becomes a codeword by flipping any one of its n bits, and it is at odd distance from every codeword. So $\rho = 1$, attained by e_1 . $\square$

For even $n = 2m$ the repetition code $\{0, \mathbf{1}\}$ has $d = 2m$, $e = m - 1$ and $\rho = m$. Every word is within $\min(\text{wt } x, n - \text{wt } x) \leq m$ of a codeword, and weight- m words are at distance m from both codewords. So it is quasi-perfect as well. This family is mentioned only for completeness and is checked in `verify.py`.

Proposition 8. *There is no binary linear quasi-perfect code of length $n \leq 1$.*

Proof. For $n = 0$ there is only one word, so no code has two codewords. For $n = 1$ the only linear code with two codewords is $\mathbb{F}_2^1$. It has $\rho = 0 < e + 1$. $\square$

Theorem 9 (length spectrum). *A binary linear quasi-perfect code of length n exists if and only if $n \geq 2$. Codes with $d = 3$ exist for every $n \geq 3$ with $n + 1 \neq 2^k$ (Theorem 5). Codes with $d = 4$ exist for every $n = 2^m$, $m \geq 2$ (Theorem 6).*

Proof. Use Propositions 7 and 8. $\square$

6 Uniform shell distributions

For a linear code C , the map $c \mapsto x + c$ is a bijection from $S_j(x)$ onto the set of weight- j words of the coset $x + C$. Also $d(x, C)$ is the minimum weight of $x + C$. Hence:

Lemma 10. *A linear code is shell-uniform if and only if any two cosets with the same minimum weight have the same weight distribution. Suppose a coordinate permutation π maps C onto itself. Then π maps the coset $x + C$ onto $\pi(x) + C$ and preserves weights. So if the group of such permutations acts transitively on the cosets of each minimum weight, C is shell-uniform.*

Theorem 11. *The following quasi-perfect codes are shell-uniform:*

1. $\text{EW}(n)$ for every $n \geq 2$;
2. $\text{EH}(m)$ for every $m \geq 2$ ($d = 4$, length 2^m);
3. $\text{SH}(2^m - 2)$ for every $m \geq 3$ ($d = 3$, length $2^m - 2$).

Proof. Identify a label $\ell < 2^m$ with the vector of its binary digits in $\mathbb{F}_2^m$. Then $\oplus$ is vector addition and $\sigma(x) = \sum_{j:x_j=1} \ell_j$.

(1) $\text{EW}(n)$ has two cosets, itself (minimum weight 0) and the odd-weight words (minimum weight 1). There is one coset of each minimum weight. (In Lean the bijection $c \mapsto c + x + y$ is given explicitly.)

(2) The coordinates of $\text{EH}(m)$ are labelled by all of $\mathbb{F}_2^m$. The map $x \mapsto (\sigma(x), \text{wt}(x) \bmod 2)$ is linear and surjective onto $\mathbb{F}_2^m \times \mathbb{F}_2$, and its kernel is $\text{EH}(m)$. So the cosets correspond to the pairs (s, p) . By the proof of Theorem 6, the minimum weights are as follows: $(0, 0)$ has minimum weight 0; $(s, 1)$ has minimum weight 1 for every s ; and $(s, 0)$ with $s \neq 0$ has minimum weight 2.

Every affine bijection $\varphi(v) = Av + b$ of $\mathbb{F}_2^m$, with $A \in GL(m, 2)$, permutes the coordinates. Let a word have support T . Its image has support $\varphi(T)$, with syndrome $A\sigma + |T|b$ and the same parity. So φ maps the coset (s, p) to $(As + pb, p)$, and it preserves $\text{EH}(m)$, which is the coset $(0, 0)$.

- The translations $(A = I, b = s + s')$ map $(s, 1)$ to $(s', 1)$.
- For nonzero s, s' , a linear map A with $As = s'$ maps $(s, 0)$ to $(s', 0)$.

So the action is transitive on the cosets of each minimum weight.

(3) Let $n = 2^m - 2$ and $u = 2^m - 1$, the all-ones vector. The labels are $\mathbb{F}_2^m \setminus \{0, u\}$. The unit vectors $1, 2, \dots, 2^{m-1}$ are labels, so σ is surjective and the cosets correspond to syndromes $s \in \mathbb{F}_2^m$. By Theorem 5 and Lemma 3, the minimum weights are: $s = 0$ gives 0; $s = u$ gives 2; and every other s (a label) gives 1. Minimum weights 0 and 2 each have a single coset.

Now let $s, s' \notin \{0, u\}$. Then $\{s, u\}$ and $\{s', u\}$ are both linearly independent, so some $A \in GL(m, 2)$ satisfies $Au = u$ and $As = s'$ (extend both pairs to bases). A fixes 0 and u , so it permutes the labels. The induced coordinate permutation multiplies syndromes by A , so it preserves the code and maps coset s to coset s' . Lemma 10 finishes the proof. $\square$

This agrees with the classical literature on uniformly packed codes (Semakov, Zinov'ev and Zaitsev; Goethals and van Tilborg), where extended and once-shortened Hamming codes are standard examples. Our proof does not depend on it. Shortened Hamming codes of other lengths are in general *not* shell-uniform. For example, in $\text{SH}(5)$ the words e_1 and e_2 are both at distance 1 from the code, but only e_1 has a codeword at distance 5, namely $e_2 + e_3 + e_4 + e_5$ (Lean: `SH5_not_shellUniform`).

7 The disproof

Corollary 12. *Clause (1) is false. In fact, already its “only if” direction fails for every finite L . This holds even when only codes with $d \geq 3$ (or $d \geq 4$) are counted, and even when perfect codes are also allowed (reading $\rho \leq e + 1$).*

Proof. Let L be finite and $M = \max(L \cup \{0\})$. Put $n = 2(M + 2)$. Then $n \geq 4$, $n \notin L$, and $n + 1$ is odd and ≥ 5 , so it is not a power of 2. By Theorem 5, $\text{SH}(n)$ is quasi-perfect with $d = 3$, but n is neither of the form $2^k - 1$ nor in L . For the $d \geq 4$ version take $n = 2^{M+2} > M$. Then $n + 1$ is odd and > 1 , and $\text{EH}(M + 2)$ is quasi-perfect with $d = 4$. A code with $\rho = e + 1$ in particular satisfies $\rho \leq e + 1$, so the non-strict reading is covered too. $\square$

Corollary 13 (uniform-shell reading). *Clause (1) is also false when “quasi-perfect” means uniform-shell quasi-perfect (Definition 2). Such codes exist exactly for $n \geq 2$, by $\text{EW}(n)$ and Proposition 8. Nontrivial ones exist for the infinitely many lengths $2^m - 2$ ($d = 3$) and 2^m ($d = 4$), and none of these lengths has the form $2^k - 1$.*

Proof. Use Theorem 11. Choose n larger than $\max L$ exactly as in Corollary 12: $n = 2(M + 2)$ for EW, $n = 2^{M+3} - 2$ for SH, or $n = 2^{M+2}$ for EH. $\square$

So the true spectrum is $\{n \geq 2\}$ under both readings, and its complement $\{0, 1\}$ is finite. The proposed “spectrum” $\{2^k - 1\} \cup L$ has density zero. The clause seems to confuse quasi-perfect codes with perfect codes. The lengths of nontrivial binary perfect codes do lie in $\{2^k - 1\} \cup \{23\}$, and the binary Golay code supplies the exceptional length.

8 Robustness of the interpretation

- **“Tight” as “nearly perfect”.** If “tight” is read in the sense of Goethals and Snover (nearly perfect codes, i.e. codes meeting the Johnson bound with equality), the conclusion is unchanged. For $e = 1$ and n even the Johnson bound reads $|C| \leq 2^n/(n+2)$, and $\text{SH}(2^m - 2)$ has $2^{n-m} = 2^n/(n+2)$ codewords, so it meets the bound. These once-shortened Hamming codes are the standard nearly perfect codes with $e = 1$ (Goethals–Snover 1972; Lindström 1975). They give the infinitely many lengths $2^m - 2$, none of the form $2^k - 1$. (On paper only.)
- **Strict versus non-strict.** Our codes have $\rho = e + 1$ exactly, so they are quasi-perfect whether or not perfect codes are counted (Lean: `onlyIfWeak_false`).
- **Trivial codes.** Someone could object to $d \leq 2$ or to repetition codes. The shortened Hamming codes ($d = 3$, every admissible $n \geq 4$) and the extended Hamming codes ($d = 4$) are standard nontrivial codes (Lean: `onlyIf3_false`, `onlyIf4_false`, `levenshteinAtLeast_false` for every bound $D \leq 4$ on d).
- **Range of k , and the “if” direction.** We refute the “only if” direction with k arbitrary. This is the weakest form, and it implies every variant that restricts k . The “if” direction is not used. (It holds for $k \geq 2$ because of $\text{EW}(2^k - 1)$, and fails only at $n \in \{0, 1\}$.)
- **The exceptional list.** L is arbitrary and finite, which includes whatever the “Levenshtein set” is meant to be. The counterexample lengths form an infinite set (all even $n \geq 4$ already suffice).
- **The conjecture’s uniform-shell notion.** This notion is handled by Definition 2, Theorem 11 and Corollary 13. In Lean, the even-weight family carries the refutation (`EW_shellUniform`, `UQP_spectrum`, `conjecture_00000007964_false_uniform`). The nontrivial uniform families $\text{SH}(2^m - 2)$ ($d = 3$) and $\text{EH}(m)$ ($d = 4$) are proved uniform in the report and checked by `verify.py`. Weaker variants that constrain only some shells are implied.
- **The first conjunct.** The “gap constant of the radius spectrum” is not defined in the conjecture. We do not address that conjunct, and the disproof does not need it.
- **Linear.** Our codes are $\mathbb{F}_2$ -subspaces. Linearity is part of the Lean definition.

9 Lean formalization

The project `lean4/` (Lean 4.19.0, core library only) defines words as `List Bool`, with weight `wt`, coordinatewise sum `xorW`, distance `dist` and the syndrome `synd i x`, the XOR of the labels $i, i + 1, \dots$ of the 1-positions. A code is a predicate `C : List Bool -> Prop`:

- `IsLinear n C`: all codewords have length n , $0 \in C$, and C is closed under `xorW`.
- `MinDist C d`: all distinct codewords are at distance $\geq d$, and some pair is at distance exactly d .
- `CovRad n C r`: every word of length n is within r of C , and some word is at distance $\geq r$ from all of C .
- `IsQP n C`: `IsLinear n C` and, for some d , `MinDist C d` and `CovRad n C ((d-1)/2+1)`.

Lemmas `minDist_unique` and `covRad_unique` show these are genuine (unique) values. The main theorems are:

- `SH_quasiPerfect`: Theorem 5 for all n . It uses `xor_top_lt` (Lemma 4, proved via `Nat.testBit`).
- `EH_quasiPerfect`: the quasi-perfect part of Theorem 6 for all $m \geq 2$.
- `EW_quasiPerfect`: Proposition 7 without the shell count. `no_QP_short`: Proposition 8.
- `hamming_not_QP`: the Hamming codes of length $2^m - 1$, $m \geq 3$, are not quasi-perfect. This shows that `IsQP` is not vacuous in the other direction.
- `QP_spectrum`: `QPLen n  $\iff$  2 <= n`.
- `conjecture_00000007964_false`: $\neg \text{LevenshteinClause}$, where `LevenshteinClause` is $\exists L : \text{List Nat}, \forall n, \text{QPLen } n \iff ((\exists k, n + 1 = 2^k) \vee n \in L)$.
- `onlyIf3_false`, `onlyIf4_false`, `onlyIf_any_false`, `onlyIfWeak_false` and `levenshteinAtLeast_false`: the variants of Corollary 12.
- Uniform-shell reading. The definitions are `DistTo C x r` (r is the distance from x to C) and `ShellUniform n C`: for any two words of length n at the same distance r from C and any j , there are maps f, g between the two j -shells with $g \circ f = \text{id}$ and $f \circ g = \text{id}$ on them. Further, `UQPLen n` means that some code satisfies both `IsQP n C` and `ShellUniform n C`. The theorems are:

- `EW_shellUniform`, for all n ;
- `SH5_not_shellUniform`, which shows the condition is a genuine restriction;
- `UQP_spectrum`: `UQPLen n  $\iff$  2 <= n`;
- `onlyIfUniform_false`, and `conjecture_00000007964_false_uniform`: $\neg \text{LevenshteinClauseUniform}$. This is the iff clause with `UQPLen`.

The project contains no `sorry`, `native_decide` or added axioms. `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`. Not in Lean: the shell-uniformity of $\text{EH}(m)$ and $\text{SH}(2^m - 2)$ (Theorem 11(2),(3)), the shell counts of Theorem 6, the repetition codes, and the dimension formula.

10 Independent computer check

`verify.py` (Python standard library) works by brute force. It lists codewords explicitly, computes d as the least nonzero weight, and computes ρ by a multi-source breadth-first search on the hypercube. It confirms:

- $\text{SH}(n)$ for $3 \leq n \leq 16$: $d = 3$, with $\rho = 2$ except at $n \in \{3, 7, 15\}$, where $\rho = 1$ (Hamming codes).
- $\text{EH}(m)$ for $m = 2, 3, 4$: $d = 4$, $\rho = 2$, and the shell counts of Theorem 6.
- $\text{EW}(n)$ for $2 \leq n \leq 14$, and the even repetition codes up to length 16.
- An exhaustive enumeration of all binary linear codes of length $n \leq 7$. There are 1, 2, 5, 16, 67, 374, 2825, 29212 of them, including $\{0\}$. Quasi-perfect codes exist exactly for $n \geq 2$ in this range. Those with $d \geq 3$ exist for $n = 4, 5, 6, 7$ (5, 15, 107 and 1260 codes).
- Shell uniformity, via coset weight distributions grouped by minimum weight: $\text{SH}(6)$, $\text{SH}(14)$, $\text{EH}(2)$, $\text{EH}(3)$, $\text{EH}(4)$ and $\text{EW}(n)$ for $n \leq 12$ are shell-uniform. Among $\text{SH}(n)$ with $3 \leq n \leq 14$, exactly $n \in \{3, 6, 7, 14\}$ are shell-uniform. $\text{SH}(5)$ is not.
- Lemma 4 for $k \leq 12$.

11 Reproduction

```
cd lean4 && lake build      # Lean 4.19.0, no Mathlib
cd .. && python3 verify.py  # about one minute
pdflatex report.tex && pdflatex report.tex
```